LSA – Week 1 Deliverables

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8 – Saad – AI

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Please note that all the questions have already been completed and filled in the collab notebook that was used to complete the week one deliverables.

1. Link to GitHub Repository: https://github.com/Saad-0013/AI-Mentorship/tree/main
2. **Agreement Rates:**

You v/s AI	**0.0267**
You v/s Gold	0.0700
AI v/s Gold	0.0267

3. **The Comparison Table**

Metric	LLM via API	Trained model (TF-IDF)
Accuracy against the benchmark	26.6% (8/30 correct)	0.86%
Time to label 30 items	~79 seconds (due to API rate limits)	5.502 seconds
Needs labelled training data?	No (Zero-shot via system prompt)	Yes (Requires the full historical dataset)
Cost per 1,000 items	Paid per token (API usage costs)	$0 (Runs locally so no additional cost involved)
Which would you ship, and why?	No. It suffered a catastrophic failure (only guessed 'Business'), is slow, and costs money at scale.	Yes. It is vastly more accurate, executes instantly, and is free to operate since we already possess the labeled data.

Figure 1: Comparison between the LLM and the Trained local ML Model

4. The Confusion Matrix

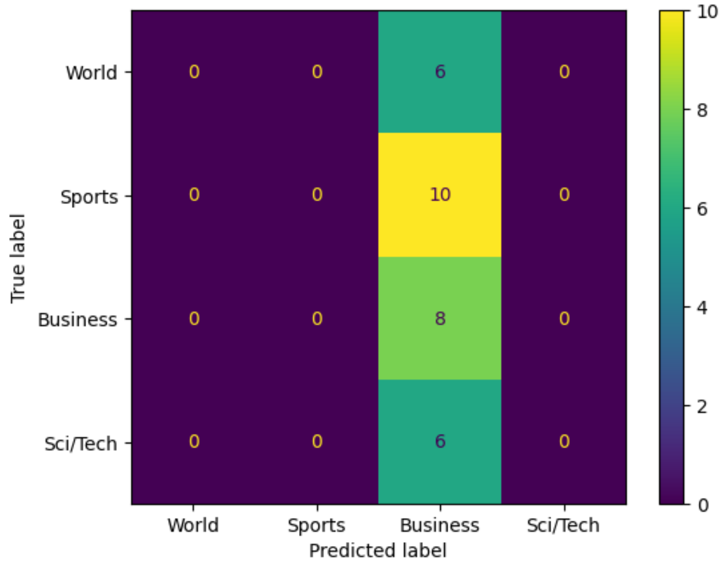

Figure 2: Confusion Matrix

Q: Which category pair was confused most often?

A: Business and Sports was the category pair that was confused the most often. It got 10 of them wrong by incorrectly labelling.

Q: Which category had the lowest recall and what that means in plain words?

A: Recall, also known as true positive rate, is how successful the model was in predicting the input correctly. World, Sports, and Sci/Tech all had the same recall which is 0%. The model failed to correctly identify even a single headline in the three previously mentioned categories.

Q: Whether accuracy alone would have told you either of these things?

A: The overall accuracy of the model was around 27%. If we had just looked at this figure we would have eventually reached to the conclusion that the model was performing poorly - and nothing else. To delve deeper into where the AI actually went wrong, we would have needed to look at additional metrics for each of the predicted values.

Skills Map:

Capability	Never done it	Done with help	Can do alone	Do it at work
Write a Python script with a loop and a function	●			
Load a CSV with pandas and filter rows			●	
Write a SQL query using a JOIN			●	
Call a web API with a key from code	●			
Use Git and GitHub		●		
Train a machine learning model on a dataset		●		
Read a confusion matrix and say what it means		●		
Write a system prompt for an LLM		●		